

Mark Solms and Michael Levin discuss explicit and implicit behavioral drives

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method: auto_caption

um yeah I wanted to get your thoughts on a on a on a specific kind of angle to some of the computational stuff that we've been doing lately sure it's always um I don't I I I I I knew that um you would have something on your mind uh but I I was busy racking my brains trying to um think what is it that we're going to be talking about and then I thought well it doesn't matter with you it's always interesting so I'll just see what it is and then we'll we'll take it from there so so what is it that's on your mind all right uh thanks um so so here's what I have um we have a um we have a pre-print and this is um Adam Goldstein and um uh attaining uh Jang a student of mine and I um have a a preprint on the following what what I wanted to do was to uh address this issue of uh nonobvious um cognitive properties in very simple systems okay and you know I I like this kind of stuff and so and so the thing that I chose specifically specifically was uh sorting algorithms so these are very simple algorithms that computer scientists of you know all CS students study these and so I'll just give you a very quick uh rundown of of what we had found and so and so they they're a good model system because they're fully deterministic so there's they're they're completely um transparent at six lines of code or so so there's nowhere to hide you know in biology there's always yet more mechanism that could be involved but here you see every all the steps of the algorithm um fully deterministic very very simple everybody knows everybody thinks they know what they're capable of everybody's been studying them it's it's very obvious and so so I thought okay let's um let's see let's see if we can find any any surprises there and and we made a couple of uh tweaks to be able to study them from this basil cognition perspective and then I think there may be sort of a psychoanalytic perspective here that I want to um that I want to probe probe with with your opinions on so the first thing we did is we just visualized um so so the way they work is you start with a string of integers let's say 100 integers they're jumbled up in random order and what the algorithms are going to do is eventually sort them into order so that

it's monotonic from you know from from one to 100 and so what you can do is you can visualize the progression of Any Given string in that space it starts off somewhere over here where the r where there it's quite random and then they all sort of move to a spot where in fact guaranteed to move to a spot where everybody's in the same order so so what you can do is you can plot this notion of sorted you know how well sorted a given string is and you see that over time it's sort of it it increases so okay so that's so that's the journey through space that this thing is taking that's the space it lives in and now we can we can ask a couple of interesting questions uh the first question that we asked was uh in the in in James's sense of um delayed gratification in other words if we put a barrier in its place is it capable of moving backwards away from its goal in order to reap gains later on you know the business with the two magnets separated by a piece of wood versus Romeo and Juliet separated and so on so um keep in mind that in these algorithms there is nothing like that there is nothing explicitly in the algorithm about well you know if you come upon a barrier then you should temporarily back off there's nothing like that we kept the algorithms exactly as they are the way we do a barrier is we simulate a broken cell so every cell has a certain number and the algorithm says move the four and the five and you know but what happens if one of them is broken they won't move if they won't move and this this this breaks an assumption in these algorithms usually you assume that the that the that the hardware is reliable if the algorithm says swap the numbers they swap and in fact the algorithm doesn't have any logic built in to see wait did it swap or didn't it it doesn't have any of that it assumes it just goes on its merry way assuming that what it said actually took place we said well biology isn't like that you have um the biological medium is is notoriously unreliable and noisy but specifically if we put a barrier you're going along and you're sorting all of a sudden you come you you need to move this number to get to to to improve things but it won't move it's broken and there one of two ways it could be broken either um it could uh uh you know refuse to be moved at all or it could refuse to um uh to to to request to be move we'll deal with that momentarily but but anyway let's say it's broken and so what we found is that um these these algorithms uh despite the fact of not having any specific code for it when they do come to a to a barrier like this they back off meaning the sordidness actually drops the string becomes less sorted they do some other stuff they move other numbers around and then eventually they go around and things and things improve and so this so they have some ability for this kind of delayed gratification they they can back away from their goal uh in order to reap gains later on and they have a little Horizon that they're willing to uh you know they're willing and able to to to do that so that was that was kind of the

first the first thing we found and that's not really the thing I want to focus on but I thought I would explain you know say it just in case you had anything interesting to say about that so so there's this there's this interesting new can I pause you for a moment um sorry you know the the I just want to be sure I'm following correctly Mike so this um algorithm uh is it a stepwise procedure or is it an end goal that it's um in other words is it aiming for the end goal the totally sorted um sequence or is it just following the steps that that are coded it's the steps there is no it is not a um it is not an explicit goal directed loop at all basically uh it it has certain criteria it says look at some numbers and then based on that shuffle the numbers and and that's it now which just so happens that if you execute that long enough you will eventually end up with a string but there is no logic in there there is no logic that asks uh is is you know how are we doing did it work how are we doing is it sorted there isn't any of that there's no self-monitoring there's no there's nothing like that yeah okay thank you so okay um the second thing we did is uh instead so so the usual these the way these algorithms are are usually done is that there's a centralized controller that's uh sort of can see Godlike can see the whole string and is following the algorithm and is moving you know moving numbers around so what we did was was we got rid of that and we said instead each cell each individual cell is following it the algorithm so the four wants a five on its on its right and the three on its left and and each each individual cell has preferences about what its neighbors are but there is no single Central controller that sees the whole thing and if you do that um it still works um and it's you know it's fine it still it still works so that's interesting you can make it sort of bottom up and distribute it in that in that way but that also allows you to do an interesting thing which you couldn't do before which is you can make a chimeric string you can make a string composed an array of numbers composed of numbers that are actually following different algorithms so so we looked at three different or three different sorting algorithms that we uh that we uh used selection sort insertion sort and bubble sort they're called so normally you have one algorithm and you're following that at one algorithm to move the numbers around but now that the numbers are moving themselves around what you can say is well some number of them are going to be following this policy some numbers some some of them are going to be following that policy and uh and if you do that it also still works meaning that they don't all have to be following the same policy but here's the here's the crazy part imagine you have a you have a an an array of of 100 cells and each of those cells has a number to it that's what you're sorting you're sorting on that number and they're starting out randomly what what we did was to to assign uh algorithms to each of those cells randomly one of two so so um Adam came up with

a nice name for it he calls it an algo type the way you have a genotype and a phenotype now now every cell has an algo type the algo type is either it's either the it's either bubble sort or it's insertion sort that's it and the algo types are also selected randomly now we ask a simple question during the lifetime of this this array as it's doing its sorting thing what are what's the internal structure like that is what is the probability that any given cell looks to its neighbor and says oh he's just like me same algo type as me what's the probability of that now in the beginning that probability is 50% because we assign algo types to numbers randomly so you know that it's a random it's a 50-50 whether the guy next to you is same as you at the end of this thing it's also random because at the end we're sorting on digits on the digit values and we know that the assignment of Digit values to algo types Is Random so at the end it's also going to be random but now the question is what does it do in the middle and what we find that it does in the middle is if if this is um this is time here and this is you know 50% it's here uh it starts out at 50% it goes up and then it then comes back down in other words what happens during the Sorting is that cells tend to Cluster next to ones like them they tend to Cluster next to their own kin basically now keep in mind in the algorithm there is nothing about there is no code in the algorithm what algo type am I what algo type is my neighbor I want to go sit next to this one I don't like sitting next to that one there's nothing like that in there if you examine the the physics of the system determined by the by the algorithm itself you don't see any of that the only time you see this is if you actually uh is is if we actually look and uh and and and and ask this question of uh we know what their algo types are and we say so so in a certain sense so you can probably see why I started thinking uh you know about analysis because it's what we are doing is uncovering a hidden cause of their behavior that is to them not Apparent at all we can see that oh you're clustering in a particular way there's a behavioral pattern that we're identifying here but it isn't explicitly coded by anything that you know if they were to examine their own code or if anybody would examine the wrong code there's nothing in the code that says that so there's so so there are a few interesting things here that I that I really like one is that it's a kind of uh it's an extremely minimal metaphor for this existential um uh condition of living things where eventually in this universe the inexorable force of the algorithm is going to pull apart all of those uh um you know clustered pieces because in the end everybody's got to get sorted according to their number So eventually the physics grinds you down and and sort of you know un undoes whatever organization you had at the beginning but during the process you have some amount of time where in a way consistent with the physics I mean there's no magic you're following the algorithm but there's

an interesting behavioral um tendency is is you know that that that you're able to satisfy for some period of time during during your lifetime and it's something that simultaneously is it's it's not against the algorithm that um that uh that you're that you're following it's consistent with it but it's also but it's not specified in the algorithm and for some amount of time it's actually at Cross purposes because because you're being you know your your behavior you're you're keeping things together that eventually are going to need to be pulled apart we also did one more experiment we asked how strong is this tendency to be to be uh to to keep together because and and one way one way you can um one way you can ask that is by allowing repeat digits if I allow repeat digits you know let's say let's say let's say there's um we we have 100 total and every and every digit can occur 10 times so there's only 10 digits total you know there's 10 fives 10 sixes and so on if you do that what happens is then you can kind of satisfy both in other words in a run of 10 fives they can be where they're supposed to be but five of them could be one algot type and five of them could be the other so it sort of makes it possible for you to retain the clustering while still be consistent with the algorithm and if you do that they cluster even more you know which which shows that these things are actually somewhat across purposes if you take off the pressure if you allow them to do what they really want to do which is to be clustered the clustering goes up um yeah so so I'm interested in this in in your thoughts on just in general this notion of of a system uh this is an extremely simple one but of course you know you and I are both interested in minimal agents and and so on um these kinds of explicit drives and behaviors that aren't in the algorithm that are emergent that are not obvious but that can be discovered if we take seriously you know uh the study of how this thing reacts in this in the problem space that it's working and if we don't assume for the moment that it's just even though it's deterministic and and so on that we don't assume that it's this dumb thing that only does what the what the algorithm says um it really you know it really it really made me feel like um yeah we were we were uncovering these hidden hidden motivations for the patterns and Its Behavior that well it's not complex enough to of course realize it but but we can see them so yeah I don't know what do you what do you think about all that um well first of all as always you know you I I I don't know um how you come up with the problems uh that that you spend your time on you know they you you have the most extraordinarily creative mind um you know so I just as as you tell the story I'm thinking this is Mike Len you nobody else is going to have this conversation with you ever um are we getting some sort of interference no I'm trying to shut off somebody's texting me and I'm trying to okay shut it off yeah um so I was thinking it through and trying to understand why um it

would do what it's doing and obviously I don't know the algorithms but it it was sort of making sense to me that why these emergent behaviors I was thinking there ways in which I can understand this would happen uh for example um your own algo type is probably more predictable uh if it's functioning by the same principle that you are you will prefer to operate in a space that is that is understandable to that behaves in the way that you expect and there's less work required from you so so you know the how that comes about um made made a sort of you know putative sense to me uh then when you ask the question um with then you know what what you were describing was of course emergent properties of the system as as you later labeled it yourself um but then you asked this very interesting question about when you say you're linking it to you're saying to psychoanalysis but really uh it's it's broader than psychoanalysis it's a A system that has explicit and implicit um principles um and so it's really the the the model I would use is is a a general sort of multi multiple memory systems sort of model and um where there are non declarative and declarative U modules and um then I was thinking uh in that the it works in the reverse uh of what you've described in the sense that what is non declarative uh is um the is the more um bottom up it's the more it's the more um sort of predetermined uh and what is declarative what is explicit uh is the more emergent you know what what couldn't be um predicted and and needs to be dealt with on the fly is what is explicit so it's an interesting uh I mean the fact that you drew that analogy which I have to say I wouldn't even have thought of again you know this is why I'm saying this is Mike Len you have to remember that you're Mark SS and that's Mike Len and now you got to start thinking differently um but since you made that connection which I wouldn't have made I find it interesting that for me it would what what I would what I would call the emergent part um is the part that's explicit um and so explicit in the cognitive sense as opposed to expit in the coding sense um you know in other words this was explicitly coded but this this Behavior emerges um what I think is explicitly coded uh in our multiple memory systems um is is what is non declarative and uh what is what what is implicit and emergent is what is declarative because that's where the uncertainty is that's where the unpredict ability is which links with my way of thinking about Consciousness which is that it has it's it's felt uncertainty it's now why do I think uh that the nondeclarative systems um are um the are equivalent to your explicit coding um I think it's as I'm trying to unpack my own intuition why do I have that intuition um well first of all I could link to what I've just said uh in that which is a kind of functionalist account uh in that um the what is implicit is what is more certain uh in the memory sense of the word in other words let me use the word n declarative so then it's clearer that I'm talking

about memory systems so the N declarative systems um are non declarative because one has a high confidence in those beliefs uh that that um they're not subject to review uh this is always the same so so in terms of a hierarchical predictive model uh you're talking about the layers in the hierarchy uh which are which which are simpler and um and more generalizable um the way I think of a predictive hierarchy by the way I don't normally picture it as a pyramid I picture it as a concentric onion uh and I think of the core of the onion as the the Deep layers uh of the predictive model and and that's where there's the greatest confidence uh and so the simplest most generalizable predictions are at the core uh and as you move to the periphery so you're dealing with with um eventualities on the fly so one of the reasons why I think of the non declarative systems um as as the um explicit in the sense of pre-programmed is because they are they are more um there's more there's more predetermination higher confidence more generalizability um and and a simpler um algorithm is being followed down there um and of course the converse applies to the periphery um the another way another reason that I think that way is because I think that the non declarative memory systems are the ones that function most like reflexes and instincts um which are um which are the pre-programmed predictions you know you these are the predictions that are determined by natural selection uh that that and and the as we automatize um our acquired predictions so they function more like reflexes and instincts they they they are more they're more automatized they're more unconscious they're more um predictable they are simpler they're more generalizable uh and the periphery of the predictive model is specialized for complexity for context sensitivity um and uh so that's where the uncertainty is um it's it's also in other words it's what is more required it is what is more um unique to to to to to me here and now as opposed to what is generally true of all versions of of of my type uh things that always function that way so um so the meaning of the word explicit and implicit uh in memory systems terms um is the opposite of the meaning of the terms uh in the way that we are using them in your in your emerging properties of your of your um of your system um so those are my uh initial thoughts um but it links to an issue that is is extremely um interesting and I think cuts to the heart of of the whole problem of Consciousness which is you know what do we mean by explicit why why does anything become explicit in the sense that the system now has to monitor its own States um I I think that is the so so I'm as you know I'm working on an artificial Consciousness project myself um and in our the the the the computational architecture that that we're um designing and adjusting as as we go along this is this has become this is one of the problems we're busy tackling right now um is is um we can make the the this our our agent function perfectly well

without there being any obvious sort of distinction between what is explicit and what is implicit and we have to kind of introduce that uh in order to try and get some what we're calling biological plausibility but it actually has two different meanings the one is to get it to function like I expect it has to you know and that's not biological plausibility it's Mark S's plausibility and the other is to make it function more like a you know vertebrate brain what what we know about how uh well actually more mammalian brains is what I'm using as my as my sort of starting point but um in in other words a brain with cortex um and um so we we're having to introduce a constraint uh you know on the way that it works in order to try to make this distinction between what is explicit and what is implicit which is of course very bad way of going about doing things because and then I realized that as we were doing that that um it's it's due to a concrete way of thinking you know of course there can be this functional distinction in the system doesn't have to actually coincide with its architecture in other words the architect you mustn't map the architecture onto your Concepts the your the concepts of what functions it's performing like what you're talking about but now these emergent properties they're not in any of the code you know that they're not any in any of the of the modules in your flow diagram but nevertheless they emerge they exist these distinctions exist um without them being coded for and without them coinciding with the with the with the brute structure of the of the algorithm or or the or the or or or the uh you know the structure of the of the the the program um but that issue of the meaning of explicit in the Consciousness sense of the word um it's it's the issue you know why does anything have to become explicit what what does explicitness do um and um it has I think to do with when a system uh is uncertain about its own state um and therefore has to focus on monitoring of its own States but nevertheless we come back to the problem uh that I started with in my response to you um which is that um it is the opposite meaning of the of the of of the of the the terms end up having diametrically opposite meaning the meaning of explicit and implicit in the phenomenological sense uh as opposed to uh in in the in the um function functional architecture sense of you know what is explicitly coded and what emerges um those are my thoughts about that problem does does that um resonate with you or lead you to any interesting places yeah super interesting I hadn't realized uh I hadn't realized about the inversion I think that's that's incredibly important and um I wonder if I wonder if tracking down when so so we can sort of look at the kinds of systems that that we're working in as like one end of the spectrum the very minimal kind and then over here is you know the sort of thing the Mamon brain you're trying to model one wonders whether that inversion takes place somewhere along that Spectrum right and whether

that's a that's some sort of face transition or whether it's a gradual Flip or I I don't know yet because I just you know I'm just taking this on now but um but yeah the fact that it's backwards uh I wonder if that is actually an important Criterion for some of the things that we are interested in you know to distinguish TR truly simple models that don't have whatever properties from okay now you're into something that's a true agent what you know maybe maybe that flip is is is important to that um and so that's the F that's the first thing so I need to think I need to think harder about this this this inverted mapping and then the other thing is that you know this this question which we wrestle with all the time in our models is like yeah what what do you bake in to make it do do whatever it's supposed to do I mean in in our example what keeps what keeps striking me is that the thing that we've observed and who knows what else right so I'm sure there's other things to observe we're just not smart enough to test for them yet you know this clustering was just something that that we thought of to do but I'm sure there's other things to do um it it doesn't have any real like we didn't have to do anything for that to emerge mean meaning it isn't it isn't any there's nothing in the algorithm that remotely is about optimizing that so to some to so to some extent this apparent uh you know the the apparent tendency of the system to optimize a particular behavior is completely emerging we didn't have to do anything and so so one wonders you know as we think about okay how do we construct things like that maybe a lot of it we don't construct maybe it it's Emer I don't know where it comes from right but but maybe it really emerges under extremely minimal conditions and that it makes me think that if something is as dumb as a as a you know six-line sorting algorithm can have these unexpected behavioral tendencies that have to be discovered by experiment that we cannot you know predict in advance the more complicated things we make you know who knows what they're doing right it's like that that's I mean that's part of why why I started this is because I wanted to have an extremely simple system yeah you know and it seems like already there you have this kind of and I I mean the thing you started with I mean it's it's amazing because it took me uh probably probably a couple of months to to to realize what you said you know right off the bat at the beginning which is that what it looks like they're doing is they're minimizing uncertainty about their neighbors right because having a neighbor who's like you means yeah so so I eventually got there too but that took me that took me a couple of months to to realize and actually Carl and I um talked about it uh a few weeks ago the thing about that and I think that's a perfectly reasonable explanation the thing about it is that there is nothing in the algorithm about that there's nothing about making predictions testing you know we're doing some we actually doing some experiments

right now to see like how many interactions they have before they cluster do they actually have to spend some time with each other before they realize that hey this is this you know this one's just like me so but but but all of that is completely emergent so I just you know this thing if it wasn't all already fundamental enough this this this surprise minimization principle it it seems now even more you know just baked in you don't need to do much of anything you all you get that you know just from from extremely minimal systems yeah um so I don't know if if there's a literature on this suddenly what you just said um made me realize that this principle um of preferring to be uh in the um neighborhood of agents that function like me because they are more predictable um they reduce my uncertainty um it it it it makes me realize with a sort of chill down my spine that this is obviously part of um of of of of prejudice you know why why people want to be with their own kind you know is that they're more predictable you you you know what's going on you know what to do you know what to expect and being with others in in the social sense uh it makes you um uncertain uh and and therefore um increases your free energy and therefore you don't want to be there it's I say it sends a chill down my spine because it makes makes me realize how you know uh how how little Prospect one has of being able to engineer um socially engineer this out of us the only the only way to do it is to make sure we all have a common enemy so but that's a sort of a side thought as I say I don't know if anybody is has has said that um maybe that's generally recognized to be the case but it for me it's a I've only discovered that in my own mind right now in this conversation yeah yeah I mean it's it's amazing that uh something like that is present at the very bottom of you know it didn't take much it didn't take any any Evolution at all in fact to come upon this right but it it actually this idea actually cropped up um about a month ago um some colleagues and I um were discussing uh you know Ai and diverse like diverse intelligence and all that and uh what we had a lot of debates about um this sort of spectrum you can imagine the sort of spectrum where on one end it's things like um objectophilia right where you you sort of completely mistake the the amount of AG agency and whatever system you're dealing with and you fall in love with a bridge or an Eiffel Tower or something you know people do of course um which which are you know sort of not at the level that that you think they are but the and so and so that's that's one type of error but but the other type of error is exactly what you just said which basically you know if you're if you're too strict about saying well I I don't believe you know you're you you don't have the um the origin story or composition that I do and therefore I don't think you have a real mind I think you're you know faking it as as what they say about certain I mean the the the far end of that is something like only love your own

kind right that's the that's that's what you were just saying you know that so so we want it so so we spent a lot of time talking about how we could arrive at principal strategies for picking criteria along that because you don't want to be at either one of these you know either one of these polls you want to um match sort of impedance match the the kind of relationship you have with a system to what and and and I firmly believe that even in the absence of you know aliens and whatnot um we're going to be surrounded by really radically different systems we're gonna have to like work on getting this right so I you know from that perspective yeah I I completely agree with you that that this this this this is an important thing to uh to get right and I think our fundamental you know that surprise minimization but I wonder you know I wonder if one thing you can do to alter how this works is to sort of re reframe what it is that you're measuring let's assume let's assume for a minute that that yes you don't want surprises you want you want to uh you know you want you want it predictable but now the question is what about and so if we can get away from superficial um kinds of differences that drive you know social you know distress and stuff like that and focus on fundamental things then then maybe this can actually make sense so so you know what I would like to be surrounded by is what you know humans with the same DNA n that's not what I'm measuring maybe we measure something else maybe we measure um kindness or pro- adaptive you know pro-social behaviors or um somebody who has the same kind of existential concerns that I do or something like that right so so so shift the measurables in a way that it's actually you know it's actually better I don't know that's that's that's that's where I've been trying to and trying to think about how to create principal uh strategies for that um so think you just put it it is very much um a concern of mine as I'm working on this artificial Consciousness uh the Prejudice uh of it's not like me therefore it can't have um this valuable thing that I've got um I'm I'm very exercised by I mean in my field I have it even with with uh colleagues who won't acknowledge Consciousness in creatures which are too far from us um in in in in terms of their phenotype um let alone you know artificial um agents the the Prejudice is just enormous and so it's interesting to link that with what we've just been talking about again I hadn't thought of that um the the solution that we um mammals at least um have come to that functions in the other direction um is an an epistemophilic um uh inclination you know that it's that it's important to um to approach what is uncertain a positive a positive attraction to to to to gaining more knowledge um and therefore to engaging with what is novel with what is not understood with what is with what is it's what becomes interesting you know so so there's a the misoc cortical misol liic dopamine system functions like that in the mamalian brain it's it's its

homeostatic set point is to be in a state of uncertainty it's it's it's it's it's a in in homeostatic deviation uh if it's bored you know if if if everything's predictable then that need is not being met the need to to engage with novelty it's a sort of what makes us forage um it's what makes us explored it's what makes people like you and me exist at all you that we we're constantly attracted to what is to what is not known um and you can see uh over longer time scales uh the biological advantage of that because to engage with uncertainty is in the long long term to reduce uncertainty um so you know the world's not going to bite you in the butt in future because you you you engaged with it uh it now proactively um but then an entirely different line of thought that was triggered in me as I was listening to what what you were saying um is that really what you're talking about is a sort of swarm intelligence isn't it uh where where each creature has a very uh um limited range of behaviors and then although each one of them is doing this very limited thing um it it it what emerges is is very complex behaviors by the by the Swarm as a whole um none of so there's no master program u in each um uh member of the Swarm but the Swarm as a whole produces this this um much more intricate um outcome that maps onto what you were talking about in terms of your algorithm uh and the interesting emergent behaviors that that that you described to me but then I realized that this links actually to one of your basic concerns which is that we multicellular creatures are all of us actually swarms of cells and um so this is uh but but we are not we are not um swarms of equal cells um we we have um we have nerve cells and and uh and bone cells and and so on um which are specialized for in some respects of course they are all of them just you know animal cells but uh in in another sense they they're not they so you do end up having sort of um m controllers in the in in in the nerve cells for example in relation to to to to other cells it's so there's a it's it's not we are not just a swarm you know we are we're we're some there's interesting ways in which you can understand the cooperation between all of our individual cells in the um on the model on the analogy of a swarm but there other ways in which that analogy breaks down then the other thought that was marked in Me by by what you were just saying um is um at what point does one is one justified in speaking of cognition I mean what is it that what does the term cognition refer to really um if a if a you have a very simple algorithm that always does exactly the same thing um where none of this kind of emergent behavior of the kind you've just described happens uh is that cognition of a very very very basil kind uh or does one only uh have a reason to speak of cognition once emergent behaviors once once there are novel problems um or at least novel phenomena uh novel comput computations which are not to go back to the very beginning of our conversation which are not

um written into the code you know computations which which are implicit in the sense that we were using which are emergent in the sense that we were using those two terms um does that qualify as cognition um and if it does bearing in mind um my own um um beliefs about what constitutes Consciousness and you you and I have spoken about this many times before functionally what constitutes a conscious what is probably definitional functionally of a conscious agent is an agent which can solve novel problems um it it has to have some way um of feeling its way through the problem in other words of monitoring is this going well or is this going badly because it's not written into my into my DNA as it were you know it's it's this is something that I've never um that that I have no predictions for I now have to I now have to predict the present uh and and and how do you do that well you have to palpate how well or badly is this going and then you have to change your mind um that's my I mean putting it in very simple simplified terms that's my Criterion for what what is U what is um plausibly potentially a conscious uh agent but but now I realize in saying that that then there's no distinction um between what qualifies as a as as something called cognition and and what qualifies as something called Consciousness um the only way I could um find a um a justification for continuing to use those words differently is that there are um operations which become automatized uh which were once not there so in other words something Rises to the level of of um explicit in the in the psychological sense of word one as of feeling your way through the problem then you learn from the experience and you lay down new predictions which can become uh very confident and automatized predictions so they're the products of Consciousness but they are not in the present functioning consciously so those I think it's it it it makes sense to me that you would have to call those cognitive operations because they're the products of learning um but the actual work of of learning learning the actual learning from experience with the emphasis on the word experience you know um that part is conscious cognition uh it's working memory as opposed to long-term memory which is which is now I laid down a prediction on the basis of the predictive work um that that that that I that I did but but let me ask you that question that I began with and what would your definition of of cognition be uh in the sense that you're looking for um basil cognition what what constitutes cognition as opposed to wrote um following a program yeah so so I agree with you I think it's fundamentally about problem solving and feeling your way through a problem and specifically in um in vulnerable creatures because uh it's different if if if you are a composite being that uh is in danger of falling apart into your component pieces it becomes a whole other business you know feeling it way through problems has a has a whole other business to it um and I think

that's that's that's what we mean by cognition and I'm willing to as you know I'm willing to take it very basily and I don't think it has to be the complex metacognitive stuff that that that we have um but what I'm realizing now uh is that I'm not sure if if there are any you know when so so so we we my initial um intuition was like yours that there would be very you know there would be rote kinds of things and then there would be these like more complex emerging things I'm not sure there are any of the first part anymore and and the reason I say that is because if if you were to look at these um algorithms how much more rote can you get they are fully deterministic there's not even a pseudo random number call there they are completely deterministic they are completely transparent you they're very simple there are there is no more hidden mechanism of biology to look at same thing with our Gene regulatory networks that we showed could have six different kinds of learning just again just couple differential equations so so I if from something as dumb and simple as this we can already see the ability to solve new problems meaning go around the barriers and right and we can already see that they are optimizing things that are nowhere explicitly provided a mechanism for like like um minimizing distinction from from your neighbor I mean I I don't know that that there are any that don't have that I mean maybe maybe we could try to come up with some but it makes me you know even if we were to come up with something extremely dumb um I I I don't know that our failure to find these kind of properties is uh anything but a lack of imagination on our part you know like like I'm you know we could come up with some sort of bit adder or something that just adds bits and we'd say Okay surely this thing doesn't do anything else other than this we can c and maybe but but but having seen this in sorting algorithms that distance has gotten really thin and now I'm starting to wonder if our inability to to observe it in simple things is just lack of imagination and our you know evolutionary training on very complicated things where we're just you know we're not good at seeing so I don't know um you know and and of course it leads me to this kind of um you know sort of a more pist view where uh the these kind of properties to me look like they're baked in at at the very beginning you know maybe wherever mathematics comes from I don't know but but but but baked in very early on and then and then the other thing I think I feel strongly about is that we have to do experiments so we cannot just observationally look at something and say well I can see it this thing is dumb it's not going to do anything I I don't I think we're not good at that at all I'm not I don't think it's possible anyway but I think we're we're just bad at it and uh we need to do experiments we need to do barriers we need to try various kinds of training techniques we need to see what is this thing doing when when we give it a little leeway you know um yeah

what what what what does it look like from its perspective and you you said you also said something very very interesting about like yeah you can you can treat this thing as a distributed agent from the sales point of view and from the sales point of view okay who are my neighbors and all of that but uh somebody in it might have been Carl or it might have been Adam but but somebody was saying the other day when we were talking about this look at it from the Clusters point of view because in forming these clusters maybe we do have a temporary high level higher level of organization there's a you know there's a and and and sure you know it's Dynamic meaning Pieces come and go but so are we we we're we're a molecular ship of thesias in the cellular one right so we we come and go to and uh maybe there's another story to tell from the perspective of each cluster as to what the world looks like so that's you know that's that's where I've been recently is is is really thinking about how to how to see the world from the perspective of these weird agents you know and what what does it look like what what is it like to be I so I so I don't think it's a crazy question to ask what's it like to to be a a chunk of data being sorted I don't you know what's it like to be the Sorting algorithm what's it like I I'm sort of now open to that that gets back to your you know the Consciousness question you know like I'm I there may be inner perspectives like valuable inner perspectives in all of these things I don't know yeah I um since our very first conversations uh I I've told you that um you you you've made me um much more radical than I when already I was in trouble uh in relation to my colleagues um in in in reducing everything to you know what the vertebrate brain stem is capable of um but um you know you've you've persuaded me that it goes way way way lower than that um in terms of what is um plausibly a cognitive and and plausibly a conscious system um I I also always end up finding that um there can't be a sharp dividing line at which point it becomes you know so there has to be uh gradation one has to speak of protoconsciousness or Proto cognition and you know and um the that that has to go all the way down you know so the the I I'm with you there it's not entirely the same as pan psychism but you know it's it's it's it's it's difficult to say in which way it's not pan psychism but now let me in trying to in trying to um to not be a pan psychist let let me take it a step further uh and it goes back to what we were saying a few minutes ago in this conversation when when when I was saying in one respect we um composite creatures are just a swarm a swarm of cells uh and each cell is just doing its you know it's from the cell's point of view it's just this is this is what I do that's what I always do there's no cognition going on here I do I do I always do exactly the same thing um and then you could say well from the cluster point of view so the where the emerging behaviors arise or from the swarm's point of view what Adam

or or Carl had said to you you know well what what is it like from the point of view of being the Swarm or what is it like from the point of view of being the cluster um that that um is one way of going but another way of going is back to the point I made um in when I said there's ways in which we are not a swarm in that we we do have Master cells um and and slave cells and they they're they're they're they're not on all fours they're not all doing the same thing once you start to have um cells within this cluster or cells within the Swarm um that are monitoring um at a higher level uh than the than the um soldiers I'm thinking of of ants now um you know once you have generals and soldiers um then you maybe maybe you you then you know can more justifiably speak of the mental processes being at the level of the of the organizing overseeing um overarching cells than than not not every cell then has equal dignity in terms of its cognitiv that that that gives me a really interesting idea that we should try experimentally in our system which is I can make some of the we can make some of the cells more powerful than others we can let them see a little bit further or we can let them push a little harder in terms of where they want to go and see what effect that has if we right right now everybody's equal but but I see your point I think it's very interesting um if there if there are regionalizations like that I wonder if we will see some sort of substructure within the I'm sort of visualizing a nervous system within the cluster right so you have a cluster and and and there are there's a pattern of of the of the more powerful cells within you know sort of organizing like making this up I don't know if this will happen or not but but I can sort of Imagine right a temporary almost nervous system I think we we'll do that I'll I'll I'll talk very it makes my mind suddenly leap back to your worms um and I remember that uh there was um I forget exactly the anatomy that you told me about but there was something about um the the uh something like the epithelial cells had functionality uh that the that the intestinal cells didn't have I forget as I say what the anatomy was but you you could you could grow a new worm from some cells and you could not grow a new worm from other cells well this is yeah the worms so the worms um they have a a population of adult stem cells and uh if you cut a piece that has even one of these stem cells in it you will get you know you can get a whole worm if you cut a piece and there are regions of the animal where you can cut off like the tip of the nose has no stem cells in it if you cut off that tip of the nose it will not you know it it won't but but I think it's um I think it's uh a matter of the hardware and not the information because uh so so these these stem cells are the source of new cell types they produce the stuff you need to make a proper worm right but I don't think they carry much of the information because we've done another experiment and this is not published yet but but we've we know what the answer

is when you when you take a little piece out of a two-headed worm and you irradiate it like crazy to kill all of the stem cells that are in it and you take just those somatic cells and you shove it into the body of a one-headed worm under some percentage of the time what ends up happening is that that one headed worm now becomes converted to be two-headed so that information did not require the stem cells it did not require new cell types the the transplant in fact will all die they you know the those cells only live 30 days or so um but the information lives on and so so I think there's a real distinction here about what you need to know what to do versus what do you need to actually build you know the body of the worm